

13. Detect missing values of a DataFrame (Display True or False)

AIM

To detect missing values in a DataFrame using Pandas.

INPUT

A DataFrame containing missing values.

CODE

```
exp13.py > ...  
1  import pandas as pd  
2  import numpy as np  
3  df = pd.DataFrame({  
4      'A': [1, 2, np.nan, 4],  
5      'B': [5, np.nan, 7, 8],  
6      'C': [9, 10, 11, 12]  
7  })  
8  print("DataFrame:")  
9  print(df)  
10 print("\nMissing Values:")  
11 print(df.isnull())
```

OUTPUT

DataFrame:				Missing Values:			
	A	B	C		A	B	C
0	1.0	5.0	9	0	False	False	False
1	2.0	NaN	10	1	False	True	False
2	NaN	7.0	11	2	True	False	False
3	4.0	8.0	12	3	False	False	False

RESULT

Thus, missing values were detected successfully using isnull().

14. Find and replace missing values in a DataFrame

AIM

To replace missing values in a DataFrame with suitable values.

INPUT

A DataFrame containing NaN values.

CODE

exp14.py > ...

```
1  import pandas as pd
2  import numpy as np
3  df = pd.DataFrame({
4      'A': [10, np.nan, 30, 40],
5      'B': [20, 25, np.nan, 45],
6      'C': [50, 60, 70, np.nan]
7  })
8  print("Original DataFrame:")
9  print(df)
10 df_filled = df.fillna(0)
11 print("\nAfter Replacing NaN:")
12 print(df_filled)
```

OUTPUT

Original DataFrame:				After Replacing NaN:			
	A	B	C		A	B	C
0	10.0	20.0	50.0	0	10.0	20.0	50.0
1	NaN	25.0	60.0	1	0.0	25.0	60.0
2	30.0	NaN	70.0	2	30.0	0.0	70.0
3	40.0	45.0	NaN	3	40.0	45.0	0.0

RESULT

Thus, missing values were successfully replaced using fillna().